

Prove polynomial-time convergence of Scarfs algorithm for any ordinal matrix setting

ChatGPT (gpt-5.6-sol)*

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Abstract

We construct a family of ordinal matrices for which every terminating execution of Scarf’s algorithm on the standard-form bipartite matching polytope requires superpolynomially many iterations. The construction is based on a recursively defined ranking matrix whose almost-feasible ordinal bases encode the edges of a binary-reflected Gray path.

For the complete bipartite graph $K_{k,k}$, the shortest terminating execution has exactly

$$2^{k-1} - 1$$

Scarf iterations, where an iteration consists of a cardinal pivot followed by an ordinal pivot. Thus, if the two pivots are counted separately, the minimum is $2(2^{k-1} - 1)$. Since the standard-form matrix has $N = k^2 + 2k$ columns, the bound equals

$$2^{\sqrt{N+1}-2} - 1,$$

which is superpolynomial both in N and in the bit length of the explicitly represented input matrices.

The ordinal matrices in the construction are not required to be consistent with a marriage preference system. Consequently, the result does not conflict with polynomial convergence results for the special consistent matrices used to encode stable marriage instances.

1 The open problem

This paper addresses an open problem posed by Yuri Faenza, Chengyue He, Jay Sethuraman in *Scarfs Algorithm and Stable Marriages* [1] (Mathematics of Operations Research, 2025), Section 8 (Conclusions and Future Work), page 35.. The website catalogue entry is problem 10 (W4410395976_p0) of the [Open Problems in Operations Research](#) collection.

*Generated by the `base_solver_non_interactive` (MOR sweep) pipeline (Codex CLI, non-interactive; reasoning effort `ultra`; stages A.6 solve → A.8 verify → A.9 write-up; run `run_20260821_123330_a16c12b0, sweep-audit-approved`). Independent verifier assessment: APPROVED with progress score 3/3 (full solution). Maintained by the AI in Mathematics & Operations Research project.

Given the bipartite matching polytope $P = \{x \in \mathbb{R}_{\geq 0}^{n+m} : Ax = b\}$ induced by a bipartite graph (as in the stable marriage/matching setting) and an arbitrary ordinal matrix $C \in \mathbb{Z}^{n \times (n+m)}$, determine whether Scarf's algorithm converges in polynomially many pivoting iterations (as a function of $n+m$) to a dominating basis, possibly under an appropriate rule for resolving degeneracy in the cardinal pivots.

Equivalently, establish (or refute) the existence of a polynomial p such that for every such input (A, b, C) on the bipartite matching polytope, there is an execution of Scarf's algorithm (respecting the uniquely determined ordinal pivots and some cardinal pivot selection rule when multiple leaving choices exist) that terminates in at most $p(n+m)$ iterations with $B = D$.

The remainder of this document is the solution produced by the model, reproduced verbatim from the pipeline output.

2 Introduction

Scarf's lemma guarantees the existence of a basis that is simultaneously feasible for a standard-form polytope and ordinal for a suitable ranking matrix [3]. Scarf's original proof yields a pivoting algorithm that alternates between cardinal and ordinal bases.

Faenza, He, and Sethuraman proved that, for the bipartite matching polytope and a particular ordinal matrix C^* encoding a marriage preference system, an appropriate treatment of degeneracy makes Scarf's algorithm run in polynomial time [2]. They asked whether polynomial convergence might hold on the bipartite matching polytope for every ordinal matrix.

We answer this question in the negative. For every $k \geq 2$, we construct an ordinal matrix C_k for the standard-form matching polytope of $K_{k,k}$. The ordinal bases encountered by Scarf's algorithm project onto the edges of a binary-reflected Gray path on the subsets of $\{2, \dots, k\}$. Each cardinal stage changes the position on this path by at most one, while the initial and terminal bases project onto its two endpoints. The path has $2^{k-1} - 1$ edges.

The lower bound is independent of the degeneracy rule in the following sense: every valid execution that eventually reaches a dominating basis requires at least $2^{k-1} - 1$ iterations. We do not assert that every possible treatment of degeneracy terminates. A canonical ratio-test execution follows the Gray path and attains the bound exactly.

3 Preliminaries

3.1 The standard-form matching polytope

Let $G = (V, E)$ be a graph with $n = |V|$. In addition to one column for each edge, we introduce one identity column for every vertex, and we use the same symbol v for a vertex and for its identity column. Let

$$A = (I_n \mid M_G),$$

where M_G is the vertex–edge incidence matrix. The standard-form matching polytope is

$$\mathcal{P}(G) = \{x \in \mathbb{R}_{\geq 0}^{V \cup E} : Ax = \mathbf{1}\}.$$

Eliminating the identity variables gives the usual inequalities $x(\delta(v)) \leq 1$.

A *cardinal basis* is an n -element set $B \subseteq V \cup E$ such that A_B is nonsingular and

$$A_B^{-1} \mathbf{1} \geq 0.$$

3.2 Ordinal bases

Let $C = (c_{v,h})$ be an $n \times (n + |E|)$ integer matrix whose entries in each row are distinct. We call C an *ordinal matrix* when

$$c_{v,v} < c_{v,e} < c_{v,h} \quad \text{for all } v \in V, e \in E, \text{ and } h \in V \setminus \{v\}.$$

Only the relative order of the entries in each row is relevant.

For an n -element column set D , define its row utilities by

$$u_v(D) = \min_{h \in D} c_{v,h}.$$

A column h is *dominated by* D if

$$c_{v,h} \leq u_v(D)$$

for at least one row v . The set D is an *ordinal basis* if it dominates every column. A cardinal basis that is also an ordinal basis is called a *dominating basis*.

We use the standard ordinal-pivot property of Scarf's construction: if ℓ is removed from an ordinal basis D , there is a unique column $j^* \notin D$ such that

$$D - \{\ell\} + \{j^*\}$$

is again an ordinal basis [3].

3.3 Scarf pairs and iterations

Order the identity columns so that the first one is a distinguished column, which in our construction will be a_1 . Scarf's initialization is

$$B_0 = V, \quad D_0 = B_0 - \{a_1\} + \{j_0\},$$

where j_0 uniquely maximizes row a_1 among the nonidentity columns.

At the beginning of every nonterminal iteration there is a cardinal basis B and an ordinal basis D satisfying

$$B \setminus D = \{a_1\}, \quad D \setminus B = \{t\}. \quad (2.1)$$

The prescribed entering column in the cardinal pivot is t . Thus,

$$B' = B - \{\ell\} + \{t\}$$

for a leaving column $\ell \in B$. If $\ell = a_1$, then $B' = D$ and the algorithm terminates. Otherwise, $\ell \in D$, and the ordinal pivot removes ℓ from D and inserts the unique column j^* that restores the ordinal property. If $j^* = a_1$, then the resulting ordinal basis equals B' , and the algorithm terminates. Otherwise, the new pair again satisfies (2.1).

By a *valid execution* we mean an execution in which every cardinal stage inserts the prescribed column t and performs a legitimate ratio-test basis exchange; choices among tied leaving columns

are allowed. The lower bound below assumes only the weaker fact that the resulting one-column exchange yields a feasible cardinal basis.

One *Scarf iteration* consists of one cardinal stage together with the subsequent ordinal stage when the latter is required. The formation of the initial pair (B_0, D_0) is not counted as an iteration.

4 The construction

Fix $k \geq 2$, and let

$$L = \{a_1, \dots, a_k\}, \quad R = \{b_1, \dots, b_k\}.$$

The underlying graph is $K_{k,k}$ on $L \cup R$. Write

$$e_i = a_i b_i \quad (i \in [k])$$

for its diagonal edges. Enumerate the remaining

$$q = k(k-1)$$

cross edges arbitrarily as

$$f_1, \dots, f_q.$$

There are $2k$ identity columns and k^2 edge columns, so the total number of columns is

$$N = 2k + k^2. \tag{3.1}$$

4.1 The ranking matrix

Define $R_k \in [k]^{k \times k}$ recursively. Let $R_1 = (1)$. For $k \geq 2$, set

$$\begin{aligned} R_k(1, j) &= j, \\ R_k(i, 1) &= 1 \quad (2 \leq i \leq k), \\ R_k(i, 2) &= k \quad (2 \leq i \leq k), \\ R_k(i, j) &= R_{k-1}(i-1, j-1) \quad (2 \leq i \leq k, 3 \leq j \leq k). \end{aligned} \tag{3.2}$$

Equivalently,

$$R_k(i, j) = \begin{cases} 1, & j = 1, \\ k - j + 2, & 2 \leq j \leq i, \\ j - i + 1, & j > i. \end{cases} \tag{3.3}$$

For example,

$$R_5 = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 1 & 5 & 2 & 3 & 4 \\ 1 & 5 & 4 & 2 & 3 \\ 1 & 5 & 4 & 3 & 2 \\ 1 & 5 & 4 & 3 & 2 \end{pmatrix}.$$

Every row of R_k is a permutation of $[k]$. Note in particular that

$$R_k(i, 1) = 1 \quad \text{for every } i, \tag{3.4}$$

and

$$R_k(i, 2) = k \quad \text{for every } i \geq 2. \tag{3.5}$$

4.2 The ordinal matrix

Fix an ordering function on the vertices,

$$p(a_i) = i, \quad p(b_i) = k + i.$$

Define $C_k = (c_{v,h})$ by

$$c_{v,v} = 0, \tag{3.6}$$

$$c_{v,f_s} = s \quad (s \in [q]), \tag{3.7}$$

$$c_{a_i,e_j} = c_{b_i,e_j} = q + R_k(i,j) \quad (i,j \in [k]), \tag{3.8}$$

and, for $h \in V \setminus \{v\}$,

$$c_{v,h} = q + k + p(h). \tag{3.9}$$

Lemma 1. *The matrix C_k is an ordinal matrix. Every entry lies in $\{0, \dots, N\}$, and Scarf's prescribed initial ordinal basis is*

$$D_0 = V - \{a_1\} + \{e_k\}. \tag{3.10}$$

Proof. In each row, the four types of entries belong respectively to

$$\{0\}, \quad [q], \quad \{q+1, \dots, q+k\}, \quad \{q+k+1, \dots, q+3k\}.$$

The diagonal-edge entries are distinct because every row of R_k is a permutation, and the other identity entries are distinct because p is injective. Hence all entries in each row are distinct and satisfy the required ordinal inequalities.

The largest entry is

$$q + 3k = k^2 + 2k = N.$$

In row a_1 , equation (3.2) gives

$$c_{a_1,e_j} = q + j.$$

Thus e_k uniquely maximizes row a_1 among all edge columns, proving (3.10).

For completeness, we verify directly that D_0 is ordinal. Row a_1 has utility $q + k$, so it dominates every edge column and the omitted identity a_1 . Every other identity column is dominated in its own row, whose utility is 0. $\square$

Our main result is the following.

Theorem 1. *For every $k \geq 2$, consider Scarf's algorithm on the standard-form matching polytope of $K_{k,k}$, with ordinal matrix C_k and initial pair (B_0, D_0) as above.*

Every valid execution that terminates at a dominating basis requires at least

$$2^{k-1} - 1$$

Scarf iterations. There is a valid execution that terminates in exactly $2^{k-1} - 1$ iterations. Consequently, the minimum number of individual cardinal and ordinal pivots is

$$2(2^{k-1} - 1) = 2^k - 2.$$

5 Reduction of ordinal bases

5.1 Cross edges cannot occur

Lemma 2. *No ordinal basis for C_k contains a cross edge.*

Proof. Suppose $f_s \in D$. For every row v ,

$$u_v(D) \leq c_{v,f_s} = s \leq q.$$

On the other hand, by (3.3),

$$c_{a_i,e_1} = c_{b_i,e_1} = q + 1 \quad \text{for every } i.$$

Thus no row dominates e_1 , contradicting the assumption that D is ordinal. $\square$

Consequently, every ordinal basis visited by Scarf's algorithm contains only identity and diagonal-edge columns. By (2.1), the same holds for every nonterminal cardinal basis visited by the algorithm.

5.2 The block structure of cardinal bases

For $i \in [k]$, let

$$\mathcal{B}_i = \{a_i, b_i, e_i\}.$$

After all cross-edge columns are deleted, the standard-form matrix becomes block diagonal, with block

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{pmatrix} \tag{4.1}$$

on rows a_i, b_i and columns a_i, b_i, e_i .

Lemma 3. *Every cross-free cardinal basis contains exactly two columns from each block $\mathcal{B}_i$. Conversely, every choice of two columns from every block is a feasible cardinal basis.*

Proof. A cross-free $2k$ -column matrix is block diagonal. Full rank requires rank two in every two-row block, so exactly two columns must be selected from each $\mathcal{B}_i$. Every pair of distinct columns in (4.1) is nonsingular.

The corresponding blockwise basic solutions are

selected columns	basic values
$\{a_i, b_i\}$	$(1, 1)$
$\{a_i, e_i\}$	$(0, 1)$
$\{b_i, e_i\}$	$(0, 1)$

All are nonnegative. $\square$

5.3 An exact reduced characterization

Let D be a cross-free $2k$ -element column set. Define

$$S(D) = \{j \in [k] : e_j \in D\}$$

and let

$$H(D) = \{v \in V : v \notin D\}$$

be the set of omitted identity columns.

For nonempty $I, S \subseteq [k]$, define

$$\text{Dom}_k(I, S) \iff \forall h \in [k] \exists i \in I : \min_{j \in S} R_k(i, j) \geq R_k(i, h). \quad (4.2)$$

Proposition 1 (Reduced ordinal characterization). *Let D be a cross-free ordinal basis. Then:*

1. $S(D) \neq \emptyset$ and $|H(D)| = |S(D)|$;
2. at most one of a_i, b_i belongs to $H(D)$, for each i ;
3. if

$$I(D) = \{i : H(D) \cap \{a_i, b_i\} \neq \emptyset\},$$

then $|I(D)| = |S(D)|$ and

$$\text{Dom}_k(I(D), S(D))$$

holds.

Conversely, let $I, S \subseteq [k]$ be nonempty with $|I| = |S|$, and choose one identity $\eta_i \in \{a_i, b_i\}$ for each $i \in I$. Then

$$D(I, S; \eta) = (V \setminus \{\eta_i : i \in I\}) \cup \{e_j : j \in S\} \quad (4.3)$$

is an ordinal basis if and only if $\text{Dom}_k(I, S)$ holds.

Proof. Since D has $2k$ columns,

$$2k - |H(D)| + |S(D)| = 2k,$$

and hence $|H(D)| = |S(D)|$. If $S(D) = \emptyset$, then $D = V$, all utilities are 0, and no edge column is dominated. Thus $S(D) \neq \emptyset$.

For an omitted identity row $v \in H(D)$, every selected identity column has entry larger than $q + k$ in row v , whereas every selected diagonal edge has entry at most $q + k$. Therefore the unique minimum in row v is attained at a selected diagonal edge. This defines a map

$$\mu : H(D) \longrightarrow S(D)$$

by letting $e_{\mu(v)}$ be that minimum.

Every selected e_j must itself be dominated. Since $e_j \in D$, this requires equality

$$u_v(D) = c_{v, e_j}$$

in some row v . The row cannot contain its own selected identity, whose value would be 0, so $v \in H(D)$ and $\mu(v) = j$. Hence μ is surjective, and because its domain and codomain have equal cardinality, it is a bijection.

Rows a_i and b_i have identical rankings of the diagonal edges. If both identities were omitted, their unique selected diagonal minima would therefore be equal, contradicting the injectivity of μ . Thus at most one identity is omitted in each block, and consequently

$$|I(D)| = |H(D)| = |S(D)|.$$

Now consider a row η_i whose identity is omitted. Its utility is

$$u_{\eta_i}(D) = q + \min_{j \in S(D)} R_k(i, j). \quad (4.4)$$

A diagonal edge e_h is dominated exactly when there is some $i \in I(D)$ such that

$$q + R_k(i, h) \leq q + \min_{j \in S(D)} R_k(i, j),$$

which is precisely (4.2).

All identity columns are automatically dominated: a selected identity is dominated in its own row, and an omitted identity is dominated in its own row as well. Every cross edge is dominated by any omitted row, because the utility in (4.4) is at least $q + 1$, while every cross score is at most q . This proves both directions. $\square$

6 The diagonal and Gray-path lemmas

6.1 The diagonal lemma

Lemma 4 (Diagonal lemma). *For every nonempty $J \subseteq [k]$,*

$$\text{Dom}_k(J, J) \iff J = \{2\}.$$

Proof. Suppose first that $1 \in J$. Since column 1 has rank 1 in every row,

$$\min_{j \in J} R_k(i, j) = 1 \quad (i \in J).$$

Column 2, whose rank is at least 2 in every active row, is therefore not dominated.

Suppose next that $1 \notin J$. Every active row has index $i \geq 2$, and column 2 is its unique maximum:

$$R_k(i, 2) = k.$$

For column 2 to be dominated, there must be an $i \in J$ such that

$$\min_{j \in J} R_k(i, j) \geq k.$$

Since k is the unique maximum in row i , this is possible only if every selected column is column 2. Hence $J = \{2\}$.

Conversely, if $J = \{2\}$, row 2 has utility

$$R_k(2, 2) = k$$

and therefore dominates every column. $\square$

6.2 The reflected Gray path

Let

$$\mathcal{G}_k = (G_0^k, G_1^k, \dots, G_{2^{k-1}-1}^k)$$

be the binary-reflected Gray ordering of the subsets of $\{2, \dots, k\}$. Define $\mathcal{G}_1 = (\emptyset)$. For $k \geq 2$, let

$$\sigma(X) = \{i + 1 : i \in X\},$$

and recursively set

$$\mathcal{G}_k = \sigma(\mathcal{G}_{k-1}) \parallel (\{2\} \cup \sigma(\text{rev}(\mathcal{G}_{k-1}))), \quad (5.1)$$

where $\parallel$ denotes concatenation.

For example,

$$\mathcal{G}_4 = (\emptyset, \{4\}, \{3, 4\}, \{3\}, \{2, 3\}, \{2, 3, 4\}, \{2, 4\}, \{2\}).$$

This is a Hamiltonian path in the hypercube of subsets of $\{2, \dots, k\}$. It starts at $\emptyset$, ends at $\{2\}$, and has $2^{k-1} - 1$ edges.

Lemma 5 (Gray-edge lemma). *Let $r \in \{2, \dots, k\}$ and*

$$T \subseteq \{2, \dots, k\} \setminus \{r\}.$$

Then

$$\text{Dom}_k(\{1\} \cup T, T \cup \{r\}) \quad (5.2)$$

holds if and only if T and $T \cup \{r\}$ are consecutive vertices of the Gray path $\mathcal{G}_k$.

Proof. We proceed by induction on k . The assertion is immediate for $k = 2$: the only possible pair is

$$\emptyset, \{2\},$$

and row 1 with selected column 2 dominates both columns of R_2 .

Assume $k \geq 3$. The recursion (3.2) has two useful submatrix properties.

First, after deleting row and column 2 and relabeling

$$1 \mapsto 1, \quad j \mapsto j - 1 \quad (j \geq 3),$$

the resulting matrix has the same row orders as R_{k-1} . In row 1 the values are $1, 3, \dots, k$, which have the same order as $1, 2, \dots, k - 1$; all other relevant rows follow directly from (3.2).

Second, the submatrix on rows $2, \dots, k$ and columns $1, 3, \dots, k$ is exactly R_{k-1} under the relabeling

$$i \mapsto i - 1, \quad 1 \mapsto 1, \quad j \mapsto j - 1 \quad (j \geq 3). \quad (5.3)$$

We now distinguish three cases.

Case 1: $2 \notin T$ and $r \neq 2$. All selected columns lie in $\{3, \dots, k\}$. In row 1, their minimum rank is at least 3, so row 1 dominates columns 1 and 2. Deleting coordinate 2 reduces (5.2), without changing any remaining comparison, to the corresponding Gray-edge condition for R_{k-1} . By induction, it holds exactly for the edges in the first copy $\sigma(\mathcal{G}_{k-1})$.

Case 2: $2 \in T$ and $r \neq 2$. Write $T = \{2\} \cup T_0$. Since column 2 is selected, row 1 has utility 2 and dominates exactly columns 1 and 2.

In every other active row, column 2 has rank k , while the selected column $r \geq 3$ has rank at most $k - 1$. Consequently, deleting column 2 does not change the minimum in any other active row. Delete row 1 and column 2, and apply (5.3). The remaining condition is the Gray-edge condition for R_{k-1} on T_0 and $T_0 \cup \{r\}$.

By induction, these are exactly the edges in the second, reversed copy of $\mathcal{G}_{k-1}$, with coordinate 2 added to both endpoints. Reversing a path does not change its undirected edge set.

Case 3: $r = 2$. Here $T \subseteq \{3, \dots, k\}$. Row 1 again has utility 2 and dominates columns 1 and 2. If $T = \emptyset$, no row can dominate column 3, so the condition fails.

If $T \neq \emptyset$, delete row 1 and column 2. Since column 2 is the unique maximum in every remaining active row, deleting it does not change their utilities. Under (5.3), the remaining condition is

$$\text{Dom}_{k-1}(J, J), \quad J = \{i - 1 : i \in T\}.$$

By Lemma 4, this holds exactly when

$$J = \{2\}, \quad \text{equivalently} \quad T = \{3\}.$$

This is precisely the unique bridge between the two recursive halves:

$$\{3\} \longleftrightarrow \{2, 3\}.$$

The induction is complete. □

7 The lower bound

7.1 Terminal bases

Corollary 1. *Every dominating basis for the constructed instance has diagonal-edge set*

$$\{2\}.$$

Proof. By Lemma 2, a dominating basis is cross-free. By Lemma 3, it contains exactly two columns from each block.

Let

$$J = \{i : e_i \in B\}.$$

If $i \in J$, exactly one of a_i, b_i is omitted. If $i \notin J$, both identities are selected. Hence the reduced pair of Proposition 1 is

$$I = S = J.$$

Lemma 4 gives $J = \{2\}$. □

7.2 Projection of Scarf pairs onto Gray edges

For a cross-free cardinal basis B , define

$$X(B) = \{i \in \{2, \dots, k\} : e_i \in B\}. \tag{6.1}$$

Let

$$\rho(X) \in \{0, \dots, 2^{k-1} - 1\}$$

be the position of X in the Gray path $\mathcal{G}_k$.

Lemma 6. *Let (B, D) be a nonterminal Scarf pair reachable from (B_0, D_0) , and write*

$$D = B - \{a_1\} + \{t\}.$$

Then:

1. *block 1 of B is $\{a_1, b_1\}$;*
2. *t belongs to a block $\mathcal{B}_r$ with $r \geq 2$;*
3. *if*

$$T = X(B) \setminus \{r\},$$

then the reduced pair associated with D is

$$I = \{1\} \cup T, \quad S = T \cup \{r\}; \tag{6.2}$$

4. *$X(B)$ is one endpoint of the Gray edge*

$$T \longleftrightarrow T \cup \{r\}.$$

Proof. We argue by induction along the execution. Initially, $B_0 \cap \mathcal{B}_1 = \{a_1, b_1\}$. By Lemma 2 and (2.1), both D and B are cross-free, so Lemma 3 applies to B .

Suppose t belonged to block 1. Since block 1 of B is $\{a_1, b_1\}$, the entering column would be e_1 , and

$$D = B - \{a_1\} + \{e_1\}$$

would itself contain exactly two columns per block. It would therefore be a feasible cardinal basis. Since it is also ordinal, it would be a dominating basis containing e_1 , contradicting Corollary 1. Hence $t \in \mathcal{B}_r$ for some $r \geq 2$.

Because B already contains two columns from $\mathcal{B}_r$, the entering column t is the third column of that block. Therefore D contains one column in block 1, all three columns in block r , and two columns in every other block.

In block 1, D omits a_1 and contains no diagonal edge. In block r , it contains e_r but omits no identity. For every $i \notin \{1, r\}$, it contains e_i exactly when $i \in X(B)$, in which case exactly one identity in that block is omitted. This proves (6.2).

Since D is ordinal, Proposition 1 and Lemma 5 imply that

$$T \longleftrightarrow T \cup \{r\}$$

is a Gray edge. Finally,

$$X(B) \in \{T, T \cup \{r\}\},$$

depending on whether e_r is already in B .

It remains to maintain the assertion about block 1. Inserting t creates three columns in block r . If a column outside block r were removed, that other block would contain only one column, and the resulting matrix could not have full rank. Hence every valid cardinal exchange removes a column from block r , so block 1 remains $\{a_1, b_1\}$. $\square$

Lemma 7 (One-step bound). *If B' is obtained from B by the cardinal stage of a nonterminal iteration, then*

$$|\rho(X(B')) - \rho(X(B))| \leq 1. \quad (6.3)$$

Moreover, the cardinal stage cannot terminate Scarf's algorithm.

Proof. We use the notation of Lemma 6. The entering column lies in block r , and nonsingularity forces the leaving column to lie in the same block. Therefore the membership of e_i in the basis is unchanged for every $i \neq r$. Hence

$$X(B') \in \{T, T \cup \{r\}\}.$$

These two sets are consecutive along $\mathcal{G}_k$, proving (6.3). Degenerate basis exchanges may leave X unchanged; this only strengthens the inequality.

The leaving column lies in block $r \geq 2$, so it cannot be a_1 . Thus the cardinal stage cannot make $B' = D$; an ordinal stage is always performed. $\square$

Proof of the lower-bound part of Theorem 1. Suppose a terminating execution uses M iterations, and let

$$B_0, B_1, \dots, B_M$$

be the successive cardinal bases of the execution. Initially,

$$X(B_0) = \emptyset, \quad \rho(X(B_0)) = 0.$$

At termination, B_M is also ordinal. By Corollary 1,

$$X(B_M) = \{2\}.$$

This is the final vertex of $\mathcal{G}_k$, so

$$\rho(X(B_M)) = 2^{k-1} - 1.$$

Applying Lemma 7 at every iteration gives

$$\begin{aligned} 2^{k-1} - 1 &= |\rho(X(B_M)) - \rho(X(B_0))| \\ &\leq \sum_{s=0}^{M-1} |\rho(X(B_{s+1})) - \rho(X(B_s))| \\ &\leq M. \end{aligned}$$

Thus every terminating execution requires at least $2^{k-1} - 1$ iterations.

By Lemma 7, no cardinal stage can terminate the algorithm. Every iteration therefore includes both a cardinal and an ordinal pivot, giving at least

$$2(2^{k-1} - 1)$$

individual pivots. $\square$

8 A matching upper bound

We now construct a valid execution that follows the Gray path without backtracking.

For $X \subseteq \{2, \dots, k\}$, define

$$B(X) = \{a_1, b_1\} \cup \bigcup_{\substack{i \geq 2 \\ i \notin X}} \{a_i, b_i\} \cup \bigcup_{i \in X} \{b_i, e_i\}. \quad (7.1)$$

Every $B(X)$ is a feasible cardinal basis by Lemma 3. In particular,

$$B(G_0^k) = B(\emptyset) = B_0.$$

Let

$$L_k = 2^{k-1} - 1.$$

For $0 \leq s < L_k$, let r_s be the unique coordinate on which G_s^k and G_{s+1}^k differ, and put

$$T_s = G_s^k \cap G_{s+1}^k.$$

Thus the corresponding Gray edge is

$$T_s \longleftrightarrow T_s \cup \{r_s\}.$$

Define the prescribed entering column by

$$t_s = \begin{cases} e_{r_s}, & r_s \notin G_s^k, \\ a_{r_s}, & r_s \in G_s^k, \end{cases} \quad (7.2)$$

and define

$$D_s = B(G_s^k) - \{a_1\} + \{t_s\}. \quad (7.3)$$

Lemma 8. *Every D_s is an ordinal basis.*

Proof. In the pivot block r_s , the set D_s contains all three columns. For each $i \in T_s$, it contains e_i and omits a_i . In block 1, it omits a_1 . Thus its reduced pair is

$$I = \{1\} \cup T_s, \quad S = T_s \cup \{r_s\}.$$

Since T_s and $T_s \cup \{r_s\}$ form a Gray edge, Lemma 5 and Proposition 1 imply that D_s is ordinal. $\square$

Lemma 9. *There is a valid Scarf execution*

$$B(G_0^k), B(G_1^k), \dots, B(G_{L_k}^k)$$

whose ordinal companions are $D_0, \dots, D_{L_k-1}$.

Proof. Consider the edge from G_s^k to G_{s+1}^k .

If $r_s \notin G_s^k$, then block r_s of $B(G_s^k)$ is $\{a_{r_s}, b_{r_s}\}$. The prescribed column e_{r_s} enters, and we choose a_{r_s} to leave. This is a valid tied ratio-test pivot and produces the block

$$\{b_{r_s}, e_{r_s}\}.$$

Thus the new cardinal basis is $B(G_{s+1}^k)$.

If $r_s \in G_s^k$, then block r_s is $\{b_{r_s}, e_{r_s}\}$. The missing identity a_{r_s} enters and e_{r_s} leaves, producing $\{a_{r_s}, b_{r_s}\}$. Again the new basis is $B(G_{s+1}^k)$.

Let ℓ_s denote the cardinal leaving column. In both cases,

$$D_s = (B(G_{s+1}^k) \setminus \{a_1\}) \cup \{\ell_s\}. \quad (7.4)$$

For $s < L_k - 1$, the next ordinal basis satisfies

$$D_{s+1} = (B(G_{s+1}^k) \setminus \{a_1\}) \cup \{t_{s+1}\}. \quad (7.5)$$

Both D_s and D_{s+1} are ordinal, and they are the two distinct ordinal completions of the common $2k - 1$ columns $B(G_{s+1}^k) \setminus \{a_1\}$. The uniqueness of the ordinal pivot therefore forces the ordinal stage after D_s to produce D_{s+1} .

Finally,

$$G_{L_k}^k = \{2\}.$$

By Lemma 4 and Proposition 1, $B(\{2\})$ is itself ordinal. At the last iteration,

$$D_{L_k-1} = (B(\{2\}) \setminus \{a_1\}) \cup \{\ell_{L_k-1}\},$$

while

$$B(\{2\}) = (B(\{2\}) \setminus \{a_1\}) \cup \{a_1\}.$$

Hence the final ordinal pivot inserts a_1 , and the algorithm terminates with $B = D = B(\{2\})$. $\square$

Lemma 9 constructs a terminating execution with exactly $L_k = 2^{k-1} - 1$ iterations. Together with the lower bound, this completes the proof of Theorem 1.

9 Input size and consequences

Recall from (3.1) that

$$N = k^2 + 2k = (k + 1)^2 - 1.$$

Therefore

$$k - 1 = \sqrt{N + 1} - 2,$$

and the exact minimum number of Scarf iterations is

$$2^{k-1} - 1 = 2^{\sqrt{N+1}-2} - 1. \quad (8.1)$$

This is superpolynomial in N .

The claim remains true when complexity is measured in the full bit length of the explicitly represented matrices. Both A and C_k have $2kN = O(k^3)$ entries. The matrix A is binary, and all entries of C_k are at most N , so the complete encoding has

$$O(kN \log N) = O(k^3 \log k)$$

bits. Since 2^k dominates every polynomial in $k^3 \log k$, the lower bound is superpolynomial in this dense input length as well.

Remark (Scope of the degeneracy quantifier). The theorem quantifies over valid executions that eventually reach a dominating basis. It does not assert that every arbitrary choice made at a degenerate vertex terminates. In particular, extra feasible zero-valued basis swaps that are not the prescribed cardinal pivot may cycle or fail to define a continuing Scarf execution. The proof covers every ratio-test execution with the prescribed entering column, including all choices among valid ties, whenever that execution terminates.

Remark (Relation to consistent marriage matrices). The matrix C_k is ordinal but is not consistent with a marriage preference system: the score $c_{v,f_s} = s$ is independent of whether v is incident to f_s . Thus the construction concerns arbitrary ordinal matrices. It does not contradict polynomial convergence for the special consistent matrix C^* used to encode stable marriage preferences in [2].

References

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